Combined and Composite Games in Evoluationary Game Theory*

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(Dated: February 23, 2026)

Building upon previous work in analyzing finite population size effects on evolutionary games, defining new game as a compositon of well known 2 and 3 strategy games and derive replicator dynamics for different microscopic interaction processes. Computing drift and drift reversal (loss of stability below a critical population size) both through extensive simulations and analytical derivations for each of the agent interaction processes. Developed visualization tools.

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Structure: You may use the `description` environment to structure your abstract; use the optional argument of the `\item` command to give the category of each item.

I. INTRODUCTION: RPS DYNAMICS

A. Evolutionary game theory

Introduction to evolutionary game theory [1, 2]

To understand the combined game, first must look at the single cases of rock paper scissors and Snowdrift.

B. Rock-Paper-Scissors Model

Examples cyclic RPS dynamics appear in biology most notably, cyclic dominance in lizards [3], strains of e-coli [4]. RPS has been analyzed in the context of evolutionary game theory extensively [2]. Previous work focusses on the rock-paper-scissors game in finite populations and has shown that a drift reversal due to finite populations can arise under certain parameter conditions and microscopic interaction processes [5].

The standard rock paper scissors matrix is defined as: The combined game analysed mainly as part of this report is a composition of the rock-paper-scissors (RPS) game and the snowdrift (SD) game.

C. Rock-Paper-Scissors Model

The general rock paper scissors payoff matrix is defined as:

$$\begin{bmatrix} a & c & b \\ b & a & c \\ c & b & a \end{bmatrix} \tag{1}$$

For the standard symmetric case of the RPS game, $a = 0, c = -1, b = 1$ where rock crushes scissors, paper covers rock, and scissors cuts paper. Ties are awarded the payoff of 0 resulting in cyclical dynamics about the internal fixed point of $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$.

1. Replicator dynamics

To analyze the time evolution of a population under the RPS game, the standard imitation based replicator equations can be derived for the system. Let x, y and z represent the fraction of players using rock, paper, and scissors respectively. The fraction of scissors players z can be defined as $z = 1 - x - y$, reducing the dimension of the replicator equations. Expected payoffs for each strategy can then be defined as follows:

$$\pi_R = ax + b\left(-x - y + 1\right) + cy \tag{2}$$

$$\pi_P = ay + bx + c\left(-x - y + 1\right) \tag{3}$$

$$\pi_S = a\left(1 - x - y\right) + by + cx \tag{4}$$

Using the standard replicator model for the evolution of population fractions over time, where the payoff of an individual is compared to the average payoff in the population, the following well known equation can be used to define a series of differential equations for each of the population fractions:

$$\dot{x}_i = x_i(\pi_i(x) - \langle \pi(x) \rangle) \tag{5}$$

where the average payoff (or mean fitness) in the population is:

$$\langle \pi(x) \rangle = x\pi_R + y\pi_P + z\pi_S \tag{6}$$

This leads to the two differential equations for the evolution of the population fractions x and y over time:

* A footnote to the article title

$$\begin{aligned}\dot{x} = &xy\left(ax - ay - bx + b\left(-x - y + 1\right) + cy - c\left(-x - y + 1\right)\right)\\ &+ x\left(-x - y + 1\right)\left(ax - a\left(-x - y + 1\right) - by + b\left(-x - y + 1\right) - cx + cy\right)\end{aligned} \tag{7}$$

$$\begin{aligned}\dot{y} = &xy\left(-ax + ay + bx - b\left(-x - y + 1\right) - cy + c\left(-x - y + 1\right)\right)\\ &+ y\left(-x - y + 1\right)\left(ay - a\left(-x - y + 1\right) + bx - by - cx + c\left(-x - y + 1\right)\right)\end{aligned} \tag{8}$$

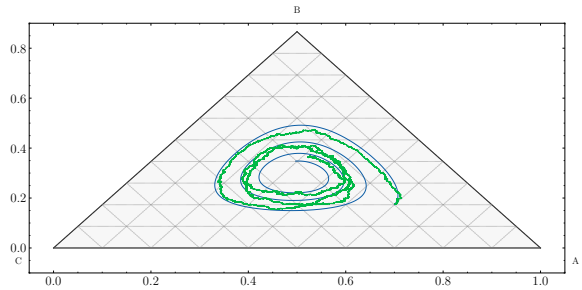

FIG. 1. Trajectory for the RPS game where $c = -0.8$. The blue line corresponds to a numerical solution of the adjusted replicator dynamics for the Moran process, and the green line is a Moran process agent based simulation with a population size of 500. Strength of selection $w = 0.3$

These are the determinisic payoff difference replicator equations from evolutionary game theory which define populations x, y over time. Later I will derive these for specific microscopic interaction processes including the noise terms that appear due to the finite population size effects.

The stability of this system of replicator equations can be analyzed by computing the eigenvalues of the Jacobian matrix of functions $F = \dot{x}, G = \dot{y}$.

In this report I will focus on the case where $a = 0, b = 1$ and c is left as a free parameter which governs the dynamics of the game. In the standard case where $c = 1$, the system is neutrally stable and the trajectories oscillate at a fixed distance from the Nash equilibrium $(1/3, 1/3, 1/3)$. When $c < -1$ the system spirals outwards from the equilibrium and will be absorbed by the boundaries and finally when $c > -1$ the system will sprial inwards, where cooperation between the 3 strategies is preferred. It has previously been shown that a drift reversal effect occurs when $c > -1$ where a critical population size can reverse the drift from the internal stable point towards to boundaries [5].

With a population size of 500, the agent based simulation follows the deterministic trajectory rather well, with added noise due to the stochastic noise element arising from a finite population size (fokker planck equation).

Compute the eigenvalues for this instance of the game and decide if its a stable point it should have real components and imaginary so it is stable equilibrium that spirals inwards.

D. Snowdrift game

The snowdrift/hawk-dove game has also been intensely studied [1] and is an example of a anti-coordination game where unlike the famous prisoners dilemma, mutual cooperation is prefered over mutual defection. It describes situations where individuals compete over limited resources and the benefits and costs associated with them.

$$\begin{bmatrix} (\frac{b}{2} - \frac{c}{2}, \frac{b}{2} - \frac{c}{2}) & (\frac{b}{2} - c, \frac{b}{2}) \\ (\frac{b}{2}, \frac{b}{2} - c) & (0, 0) \end{bmatrix} \tag{9}$$

The snowdrift game describes a scenario of two drivers trapped in a snowdrift. They have the options to shovel the snow an clear a path (cooperate) or stay in their car (defect). If both cooperate then they pay a shared cost $\frac{c}{2}$ but share the full benefit and can both drive. However the best outcome for one player is to remain in their car and let the other do all the work in turn leaving the full cost c to the other player but if both players remain in their car then both remain stranded and receive a payoff of 0. This scenario is often arises in biology and social/economic settings where mutual cooperation would be beneficial and both parties refusing to yield results in the worst outcome. For example in negotiation scenarios/research investing for competing companies/animals and microbes sharing resources in yeast cells [6] where some cooperating cells produce a necessary enzyme meanwhile "cheater" cells do not yet still receive the benefit they provide, other examples of cheating microbes [7].

Both the snowdrift and the hawk-dove game model a scenario where two players are competing over some resource but in a social/economic scenario and a biological competition however in both the cost of cooperation is high, and the cost of mutual defection is devastating. The resulting Nash equilibrium are two pure strategy equilibrium where players select opposite strategies (one gets the whole reward and no cost, whilst the other takes a smaller reward but incurs a heavy cost), and the mixed strategy $(\frac{1}{2}, \frac{1}{2})$ between cooperate and defect, where players randomize their selection as defecting all the time results in a payoff of 0.

The mixed strategy case is typically stable [show plot and calculate eigenvalues for a typical example of the snowdrift game.]

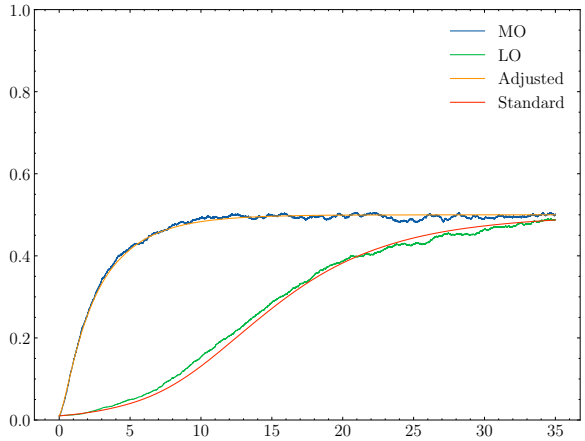

FIG. 2. Snowdrift game population time series for Local update (LU) and Moran process (MO), alongide solution to determinisic replicator equation for each process from the adjusted dynamics (ref fokker planck equation). $N = 10000$, large population size trajectories closely follow deterministic path. $w = 0.9$, initial population 99% cooperators. The game fixes at the mixed strategy Nash equilibrium $(\frac{1}{2}, \frac{1}{2})$

II. COMBINED GAME DEFINITION

This work focusses on the composite game with the idea of combing snowdrift (SD) and rock-paper-scissors (RPS) into a single four strategy payoff matrix. Average payoffs, replicator equations and fixed points can subsequently be derived and the deterministic dynamics can be studied. The simulation tool can do all of this, plot trajectories both in 2d and and in the 3d pyramid simplex.

The combined game is defined as:

$$\begin{bmatrix} a & c & b & \gamma \\ b & a & c & \gamma \\ c & b & a & \gamma \\ a+\beta & a+\beta & a+\beta & 0 \end{bmatrix} \tag{10}$$

where the values a, b, c represent the RPS cyclic game.

When $a = 0$, standard RPS ties, this can be seen in the 2x2 snowdrift context as the following payoff matrix:

$$\begin{bmatrix} \langle RPS \rangle & \gamma \\ \beta & 0 \end{bmatrix}$$

This is what the value H_{SD} will essentially be analyzing.

The population can be visualized as a 3d project of a 4-simplex with RPS at the base and the fourth (SD) strategy at the top. Cooperation in this game essentially collapses into playing RPS, whilst defection/cooperation pairs lead to typical snowdrift dynamics.

A. Replicator dynamics

Replicator analysis in the infinite population case using typical replicator dyanmics [2].

B. Stability analysis

Stability of the system, an example with interior coexistence with a stable mixed strategy SD game and cyclical RPS with a diagram. Show eigenvalues.

III. INTERACTION PROCESSES

To study the dynamics in finite populations, an interaction process is required for comparison between selected agents at each time step.

A generalized form of the Moran process in terms of transition probabilites between states in the game simplex has been discussed in Ref.[8] and I have implemented in the simulation code for this project for any number of strategies (in this work focussing on 4 strategy model). Later these transition probabilities can be used to derive a master equation [9] which can be expanded in a Taylor series to derive an equation in the form of a Fokker-Planck equation describing the time evolution of the probability densities of the population states.

IV. MODEL AND STOCHASTIC DYNAMICS

We consider a finite population of size N evolving under stochastic evolutionary dynamics with S strategies. The population state is described by the vector $\mathbf{x} = (x_1, \ldots, x_{S-1})$, where $x_i = n_i/N$ and the final strategy is fixed by normalization. At each update step, individuals interact according to a payoff matrix $\mathbf{A}$ and reproduce according to a specified microscopic update rule. In the large–N limit, the stochastic dynamics can be approximated by a Fokker–Planck equation governing the probability density $P(\mathbf{x}, t)$.

V. FOKKER-PLANCK EQUATION

Master equation [9] can be derived and using the 2nd order taylor expansion of the transition probabilities for each strategy, a fokker-planck equation can be derived for each of the processes.

In the diffusion approximation, the master equation may be expanded to second order in $1/N$, yielding a Fokker-Planck equation of the form

$$\frac{d}{dt}P^{\tau}(x) = \frac{1}{N}\left[-\frac{d}{dx}(a(x)P^{\tau}(x)) + \frac{1}{2}\frac{d^2}{dx^2}\left(b^2(x)P^{\tau}(x)\right)\right] \tag{11}$$

Here $x \in (i, j, k)$ denotes the frequency of a given strategy in the RPS subsystem, and

$$a(i) = T^{PR} + T^{SR} + T^{+R} - T^{RP} - T^{RS} - T^{R+}$$

$$b(i) = \sqrt{\frac{1}{N}\left(T^{PR} + T^{SR} + T^{+R} + T^{RP} + T^{RS} + T^{R+}\right)}$$

for strategy i, rock and similarly $a(j), b(j)$ for paper and $a(k), b(k)$ for scissors can be defined in the same way with the pairs of transition probabilities into and out of the respective states and must be derived for each microscopic update process and set of payoff matrices. Full derivation of Fokker-Planck equation and expressions for $a(x), b(x)$ can be found in full report detailing 2nd order expansion of the master equation for the 4x4 game.

For each strategy, $a(x)$ corresponds to the deterministic replicator equation ($N \to \infty$), and $b(x)$ is the diffusion term corresponding to the stochastic drift. Code is capable of computing these values for each interaction process in order to plot the correct determinisic adjusted dynamics. For the local update process, this $a(x)$ is equal to the standard replicator payoff difference equation (with some scaling factor accounting for selection pressure w).

The standard replicator dynamics [2] correspond to the local update process, and matches the replicator equation derived from FP expansion, whereas the Moran process corresponds to the dynamics proposed by John Maynard Smith [1], where there is an additional normalization by the average payoff of the population. Maynard discussed whether the standard dynamics were too restrictive for cases where fitness is frequency dependent (in a finite population), and therefor proposed the adjusted dynamics where you can imagine an individual playing against the population as a whole via the average payoff term.

VI. TRAJECTORIES

This section will contain both numerical trajectories from the fokker p equation, and the simulated results from simulation engine. Also the link to all three processes and their deterministic limit in infinite N.

VII. DRIFT MEASURES

A measure of the drift from the internal fixed point of this system can be defined by an observation function H [5, 10, 11], and approximating the value of $\dot{H}$ with a discrete sum over the population. Previous work in Ref. [5, 10, 11] has examined this both the 2 and 3 strategy case where only one observable H is used, however to analyze the combined game an observable will be defined for each of the sub games to properly describe the dynamics and drift. These will be H_{SD}, H_{RPS}, and H_4 as approximation of the distance from the SD equilibrium , the RPS equilibrium and then the overall central equilibrium as a whole respectively.

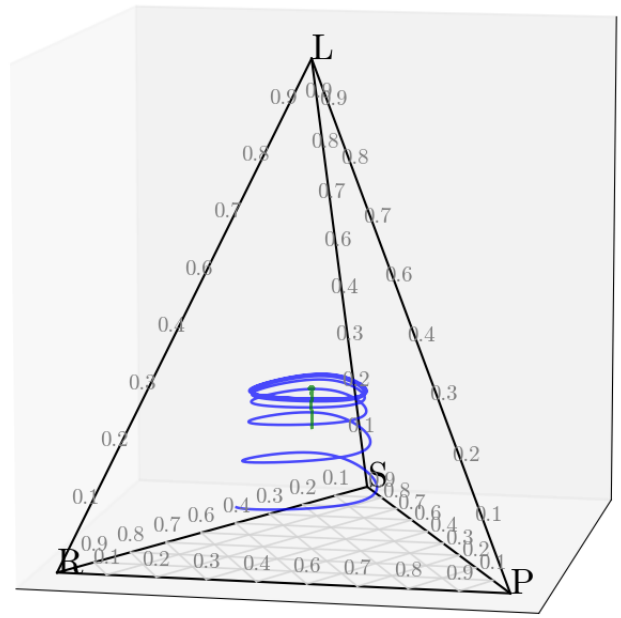

FIG. 3. Example of a trajectory with neutrally stable RPS component. The blue trajectcory corresponds to the numerically solved adjusted dynamics for the Moran process and the given payoffs. The interior green trajectory is an average trajectory for a population of 2000, averaged over 1000 independent simulation runs to show the general drift of the population in the SD component of the game. Note that drift is agreeing with replicator dynamics as $N_c < 2000$

An approximation for $\dot{H}$, $\langle \Delta H \rangle$ can be computed directly from the agent based simulation, however for all the interaction processes the terms can be formulated as an integral and solved numerically on in some cases e.g. Local update, can be solved analytically. This is preferred over simulated approximate values as it is much simpler to compute particularly in the local update case where an exact result can be found for $\langle \Delta H \rangle$. [Put the graph here of critical N for simulated vs exact critical N.]

For each of the components making up the combined 4x4 game a suitable observation value H can be defined to measure the distance from the Nash equilibrium [5, 11]. The sign of $\dot{H}$ can therefore indicate whether the system is drifts towards or away from the interior equilibrium [11]. $\dot{H}$ is approximated in the simulated results by $\langle \Delta H \rangle$:

$$\langle \Delta H \rangle = \mathbb{E}[H(\mathbf{x}_{t+1}) - H(\mathbf{x}_t)]. \tag{12}$$

which is calculated for one single update step, and then averaged over a very large number of repeated simulations. Drift reversal is defined as the change in sign of this value under variation of parameters of the system most notably the population size N.

The sign of $\langle \Delta H \rangle$ determines whether stochastic drift favors interior coexistence or boundary attraction. $\langle \Delta H \rangle$

can be formulated analytically [5] and written as a sum over the simplex, in this case: $(1 \leq i, j \leq N-i, k \leq N-i-j)$.

$$\langle \Delta H \rangle = \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} (H_s - H_{s'})\, T^{s \to s'} \tag{13}$$

for all the states $s, s' \in \{R, P, S, L\}$ and transition probabilities $T^{s \to s'}$. Where transition probabilites $T^{s \to s'}$ must be defined for each of the microscopic interaction processes. In the combined SD, RPS game there are 3 observation values for drift, H_{SD} for effects within the snowdrift game, H_{RPS} for the RPS subgame and H_4 for the overall drift in the context of the internal fixed point of all 4 strategies in the payoff matrix.

After defining the summation in terms of discrete population counts for each of the strategies, it is possible to approximate these values in the continuous limit by replacing the counts with their corresponding fraction of the population and the term is converted to an integral over the simplex allowing for the use of numerical methods to compute these exact values of $\langle \Delta H \rangle$, and therefore compute numerically the critical population sizes N where the dynamics shift and drift reversal occurs (if it does occur). This allows for more efficient comparisons of the dynamics of a particular instance of the game parameters as it is much more efficient to compute this equation numerically than to simulate over a large number of realizations and average the discrete results.

A. Simulated results

B. Rock-Paper-Scissors Drift

For the three-strategy subsystem, the replicator equations have the constant of motion:

$$H_{\text{RPS}} = -xyz \tag{14}$$

$$\begin{aligned}
\langle \Delta H_{\text{RPS}} \rangle = \frac{2}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\
\Big[z(y-x)(T^{RP} - T^{PR}) \\
+ y(z-x)(T^{RS} - T^{SR}) \\
+ x(y-z)(T^{SP} - T^{PS}) \\
+ xz(T^{P+} - T^{+P}) + yz(T^{R+} - T^{+R}) \\
+ xy(T^{S+} - T^{+S}) \Big] \\
+ \frac{2}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\
\Big[z(T^{RP} + T^{PR}) + y(T^{RS} + T^{SR}) \\
+ x(T^{SP} + T^{PS}) \Big]
\end{aligned} \tag{15}$$

This expression shows that finite population drift is controlled by antisymmetric differences of transition probabilities between competing strategies.

C. Snowdrift Drift

The 4th strategy occurs with frequency $q = 1-x-y-z$, we define the snowdrift coexistence measure

$$H_{SD} = -q(1-q) \tag{16}$$

The expected value of ΔH_{SD} over the entire simplex can be defined, similarly to the RPS case, and approximated by a continuous integral as follows:

$$\langle \Delta H_{\text{SD}} \rangle = \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \tag{17}$$

$$\Big[\Big]$$

$$\begin{aligned}
\langle \Delta H_{\text{SD}} \rangle = \frac{12}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\
\Big[q \big[(T^{R+} - T^{+R}) + (T^{P+} - T^{+P}) \\
+ (T^{S+} - T^{+S}) \big] \Big] \\
+ \frac{12}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \\
\Big[T^{+R} + T^{+P} + T^{+S} \Big]
\end{aligned} \tag{18}$$

Since transitions purely within the RPS subgame do not change the value of H_{SD}, this exact term for depends only on transitions into and out of the fourth strategy $(+)$.

Consider the case of $(H_{SD} - H'_{SD})T^{RP}$ in the summation being zero, since transitions between rock and paper do not change the frequency of the fourth strategy, and therefore do not change the value of $-q(1-q)$.

This results in $H_{SD} - H'_{SD} = H_{SD} - H_{SD} = 0$ for all transitions $(R \to P, P \to S, S \to R$, and their reverses) within the RPS subgame. For full derivation see the full report.

D. Four-Strategy Coexistence Drift

To capture full four-strategy coexistence, the constant of motion H_4 is defined

$$H_4 = -xyzq \tag{19}$$

which vanishes on all faces of the simplex and is maximized in the interior.

The expected value of ΔH_4 can be defined as the sum over all the states below

$$\begin{aligned}\langle \Delta H_4 \rangle = \frac{6}{N^7} \sum_{i=1}^{N} \sum_{j=1}^{N-i} \sum_{k=1}^{N-i-j} \Big[& ijk(N-i-j-k)(T^{RP} + T^{RS} + T^{R+} + T^{PR} + T^{PS} + T^{P+} \\ & + T^{SR} + T^{SP} + T^{S+} + T^{+R} + T^{+P} + T^{+S}) - (i-1)(j+1)k(N-i-j-k)T^{RP} \\ & - (i-1)j(k+1)(N-i-j-k)T^{RS} - (i-1)jk(N-i-j-k+1)T^{R+} \\ & - (i+1)(j-1)k(N-i-j-k)T^{PR} - i(j-1)(k+1)(N-i-j-k)T^{PS} \\ & - i(j-1)k(N-i-j-k+1)T^{P+} - (i+1)j(k-1)(N-i-j-k)T^{SR} \\ & - i(j+1)(k-1)(N-i-j-k)T^{SP} - ij(k-1)(N-i-j-k+1)T^{S+} \\ & - (i+1)jk(N-1-i-j-k)T^{+R} - i(j+1)k(N-1-i-j-k)T^{+P} - ij(k+1)(N-1-i-j-k)T^{+S} \Big]\end{aligned} \tag{20}$$

In the continuous limit

$$\begin{aligned}\langle \Delta H_4 \rangle = & \frac{6}{N} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[zq(y-x)(T^{RP} - T^{PR}) + yq(z-x)(T^{RS} - T^{SR}) + xq(y-z)(T^{SP} - T^{PS}) \\ & + yz(q-x)(T^{R+} - T^{+R}) + xz(q-y)(T^{P+} - T^{+P}) + xy(q-z)(T^{S+} - T^{+S}) \Big] \\ & + \frac{6}{N^2} \int_0^1 dx \int_0^{1-x} dy \int_0^{1-x-y} dz \Big[zq(T^{RP} + T^{PR}) + yq(T^{RS} + T^{SR}) + xq(T^{SP} + T^{PS}) \\ & + yz(T^{R+} + T^{+R}) + xz(T^{P+} + T^{+P}) + xy(T^{S+} + T^{+S}) \Big]\end{aligned} \tag{21}$$

This expression naturally decomposes into RPS–like drift terms, snowdrift–like drift terms, and diffusion corrections. The sign of $\langle \Delta H_4 \rangle$ determines whether stochastic dynamics favors four–strategy coexistence or boundary attraction.

E. Weak selection limit

Value of these integrals as $w \to 0$

F. Local update analytical solutions

For the Local update process, an exact solution for the values $\langle \Delta H_{SD} \rangle$ and $\langle \Delta H_{RPS} \rangle$ can be found.

Considering the payoff matrix:

$$\begin{bmatrix} 0 & c & 1 & \gamma \\ 1 & 0 & c & \gamma \\ c & 1 & 0 & \gamma \\ \beta & \beta & \beta & 0 \end{bmatrix} \tag{22}$$

In this case the values of a, b have been set to 0 and 1. The value of c can be altered to adjust the dynamics of the RPS component of the game, if $c > -1$ then the game will spiral inwards towards a central fixed point and if $c > -1$ the RPS trajectories will sprial outwards as shown in the introductory section. The values of β and γ determine the dynamics of the snowdrift component of the game with the main cases being fixation at either the top (pure 4th strategy), base (pure RPS) or a central equilibrium. This depends on the relative payoff difference between the SD and RPS game.

Quick derivation and explanation of the H_{SD} analytical result.

G. Comparison of drift measures

Drift reversal, the loss of stability of an attracting fixed point (something about hopf bifurcation) below a critical population size, occurs at the point where the value of $\langle H \rangle$ changes sign. Depending on the initial parameters, this effect is observed in the RPS and SD subgames, and in some cases in both.

In the case where $N'_{RPS} < N < N'_{SD}$ the point cloud of trajectories from randomly initialized simulations will be attracted to the equilibrium of the RPS game, but will exhibit a drift reversal in the SD game, seen in the SD reversal case in Fig. 4.

$N'_{SD} < N < N'_{RPS}$ disk shape

$N < N'_{SD}, N'_{RPS}$ random cloud / absorbed by all boundaries

VIII. INTERPRETATION

While $\langle \Delta H_{\mathrm{RPS}} \rangle$ and $\langle \Delta H_{\mathrm{SD}} \rangle$ describe drift within lower–dimensional subspaces, $\langle \Delta H_4 \rangle$ captures stability of full interior coexistence. Sign reversals of $\langle \Delta H_4 \rangle$ generally occur at larger population sizes, reflecting the additional geometric suppression of interior states in the four–strategy simplex.

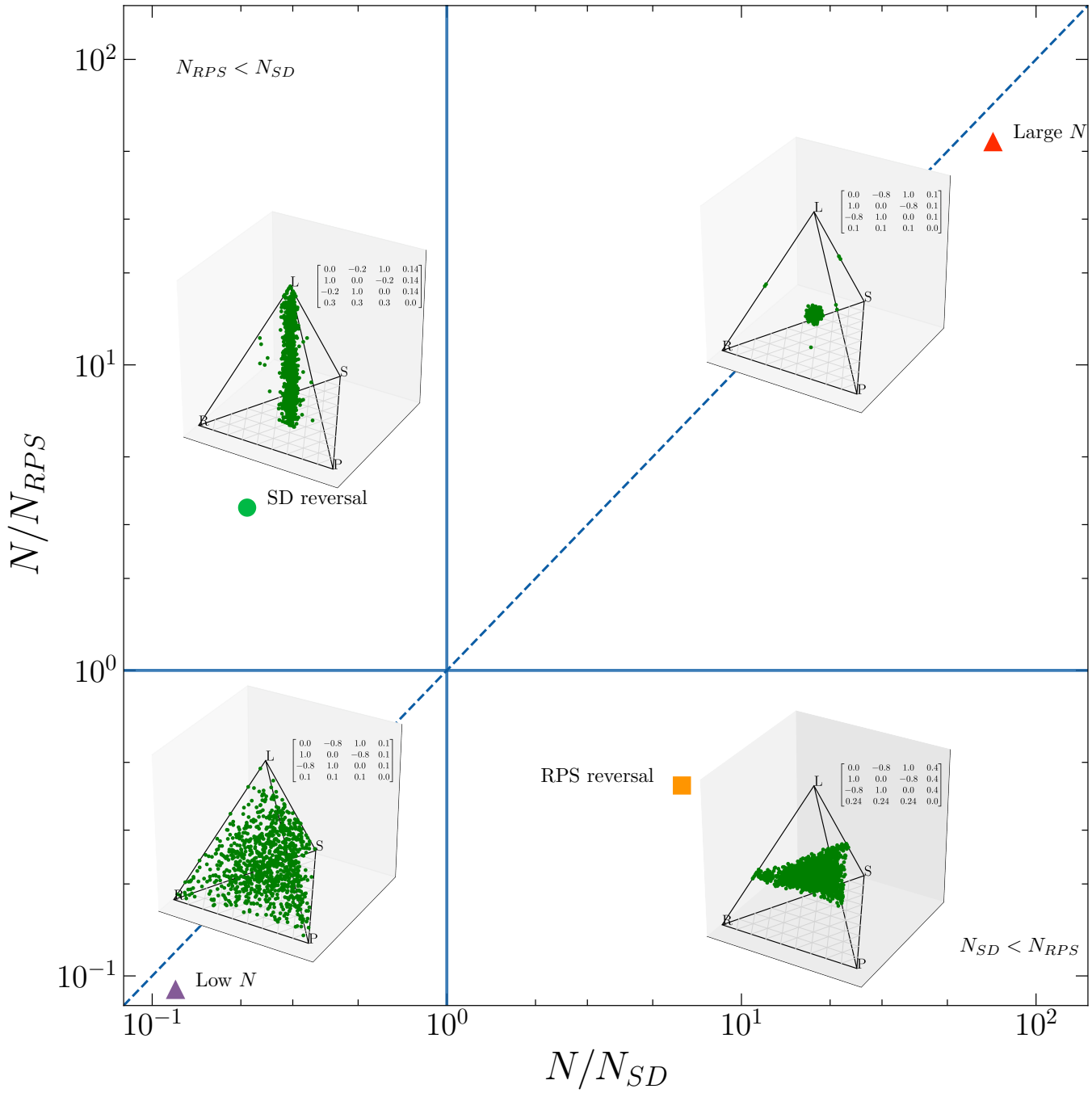

FIG. 4. Comparison of the main drift reversal cases, notably the RPS and not SD reversal (disk like shape), SD and not RPS reversal (column shape), and large N Nash equilibrium stable case where $N >> N_c$ and chaotic low N case (drift in all directions) where $N << N_{RPS}, N << N_{SD}$

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